

Environment drives residue-level adaptations while phylogeny dictates the biochemical backbone of ESCRT proteins in Asgard archaea



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Eukaryogenesis, Membrane compartmentalization

Eukaryotes
ESCRTs: CHMPs, VPS4

Asgard
ESCRTs: CHMP 1-3, CHMP 4-7, VPS4

Bacteria
ESCRTs: Vipp1, PspA

Archaea
ESCRTs: CdvB, CdvC

ESCRT III

VPS4

VPS4A

VPS4B

VTA1

Cytosol

Lumen

ILV

Introduction & hypothesis

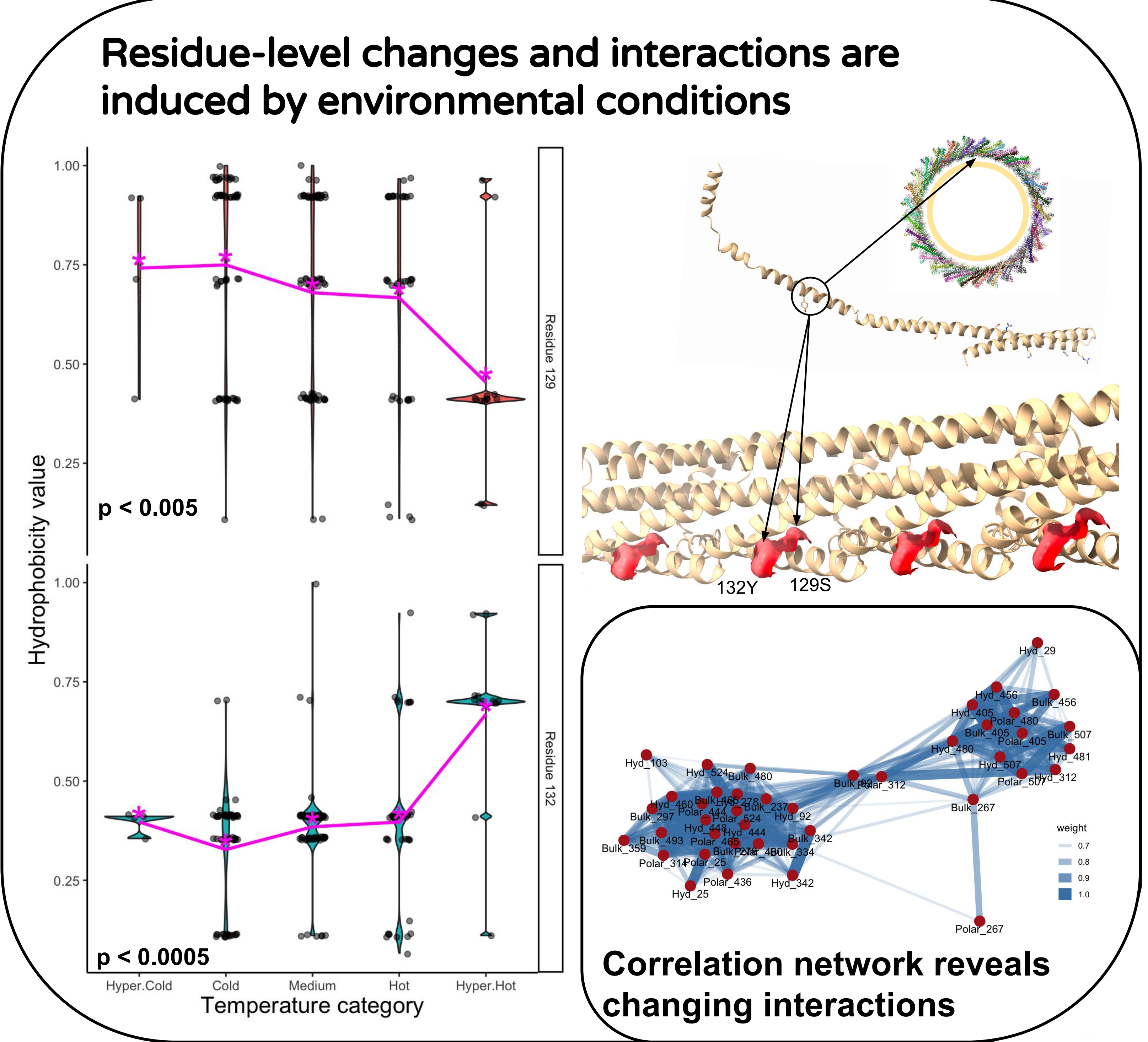
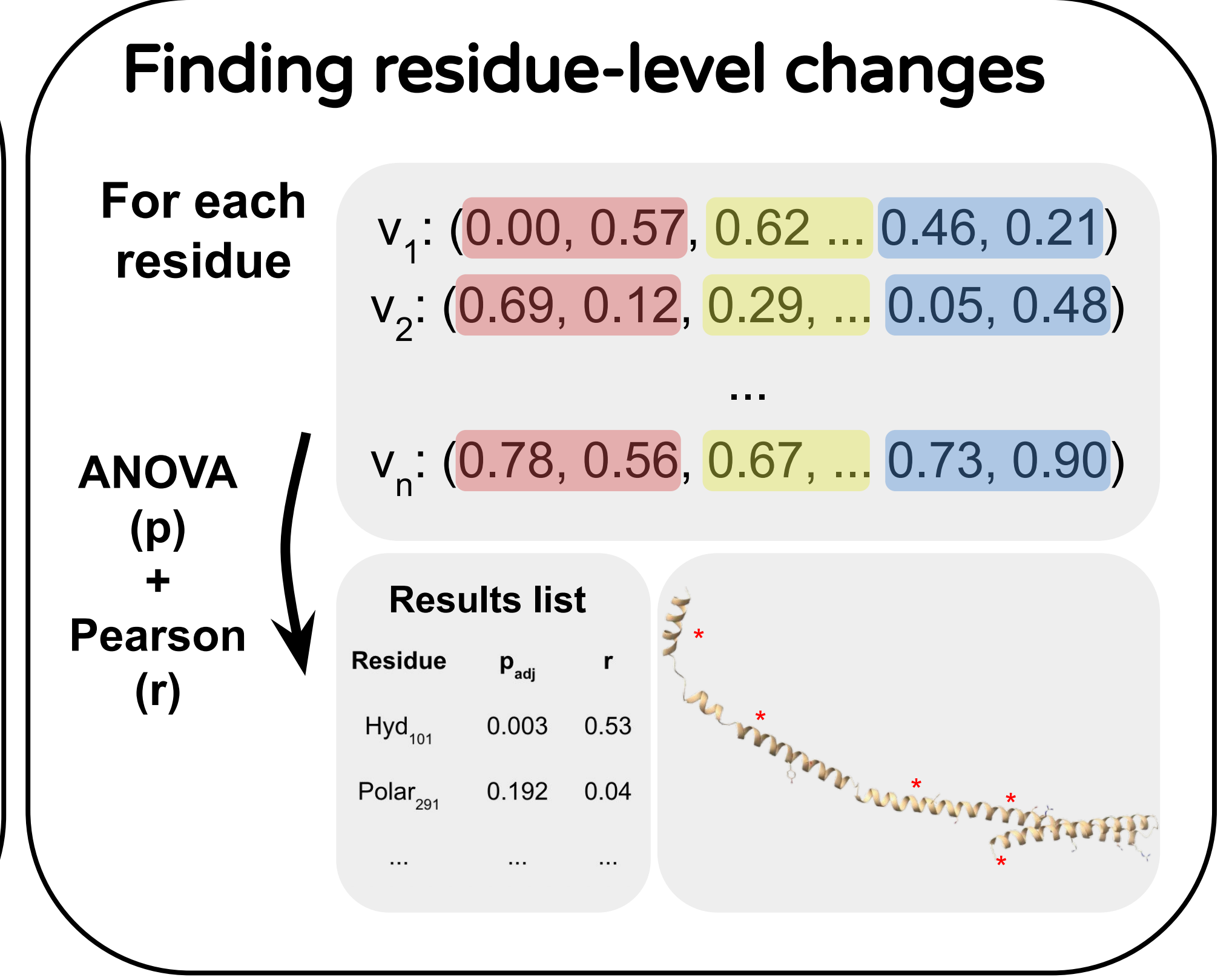
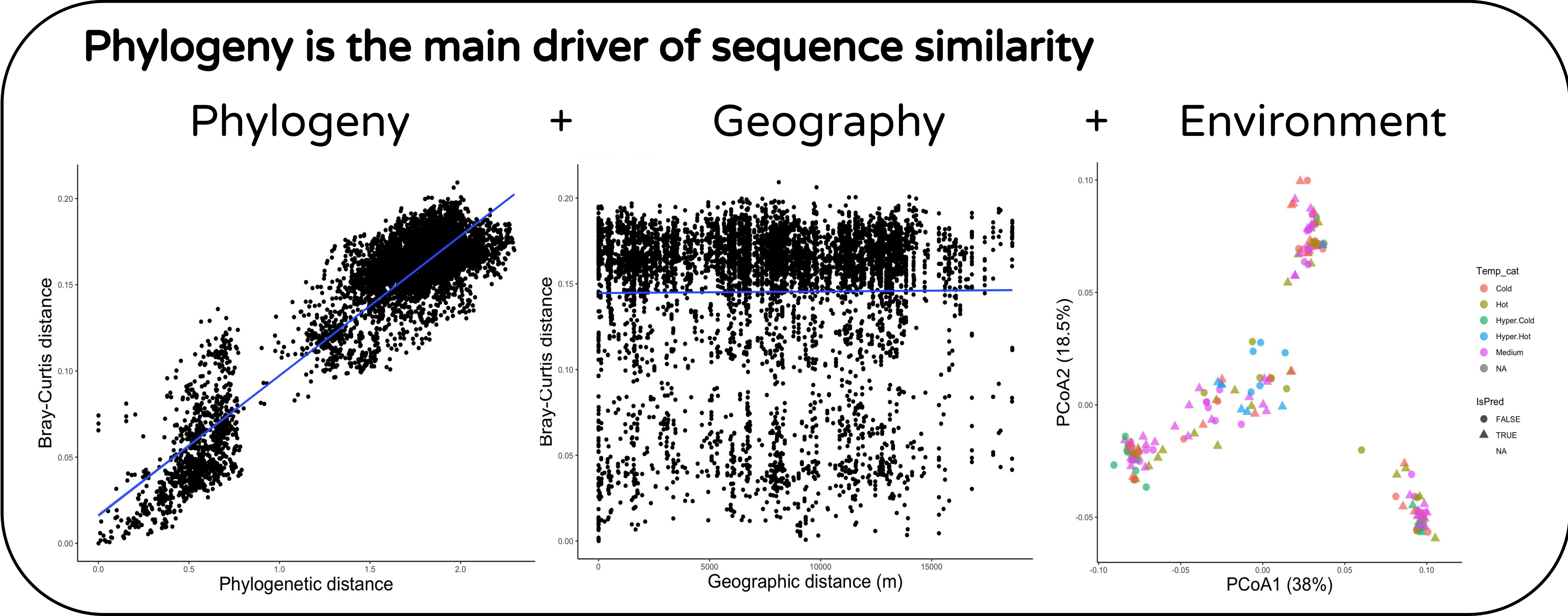
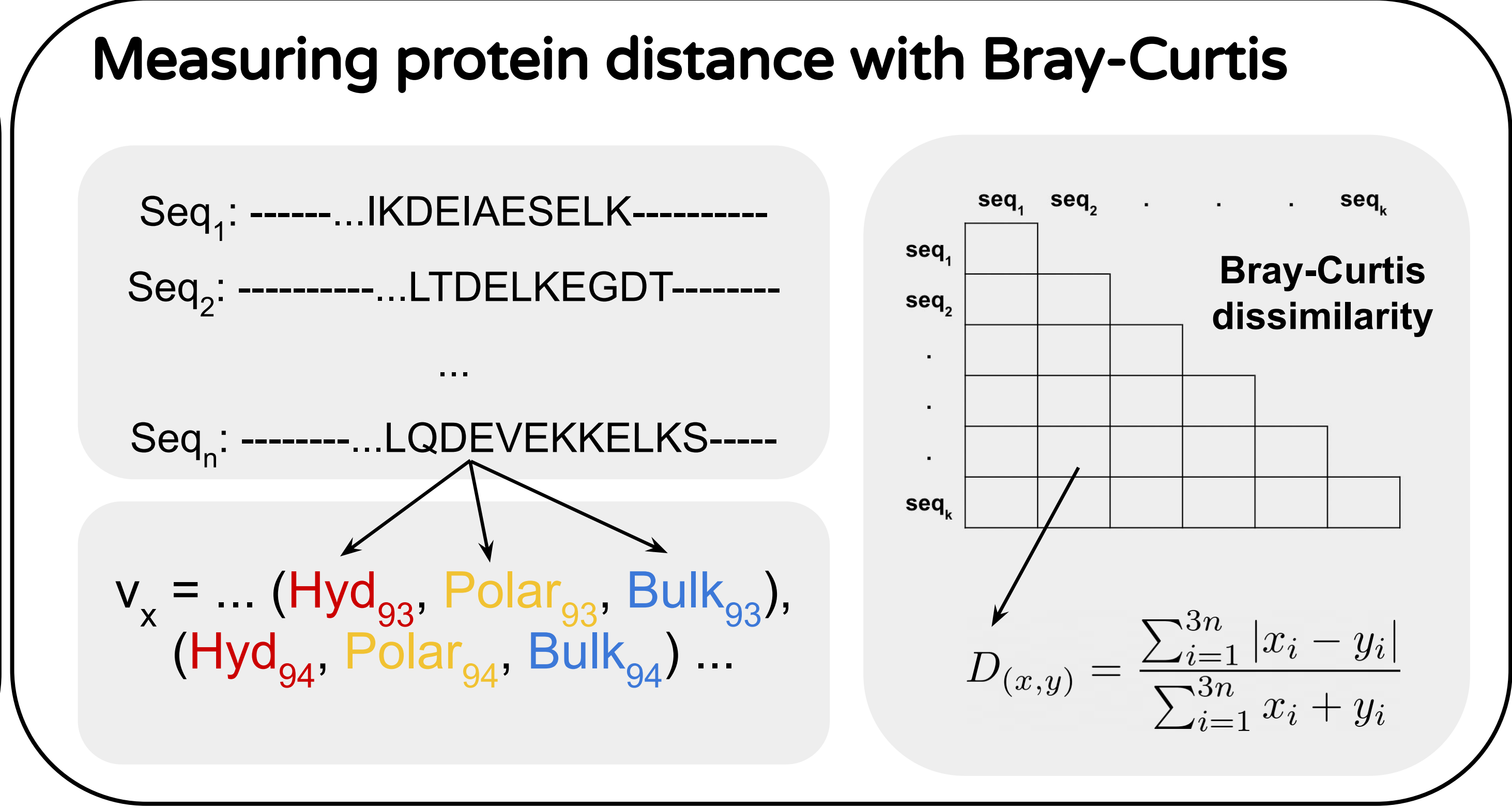
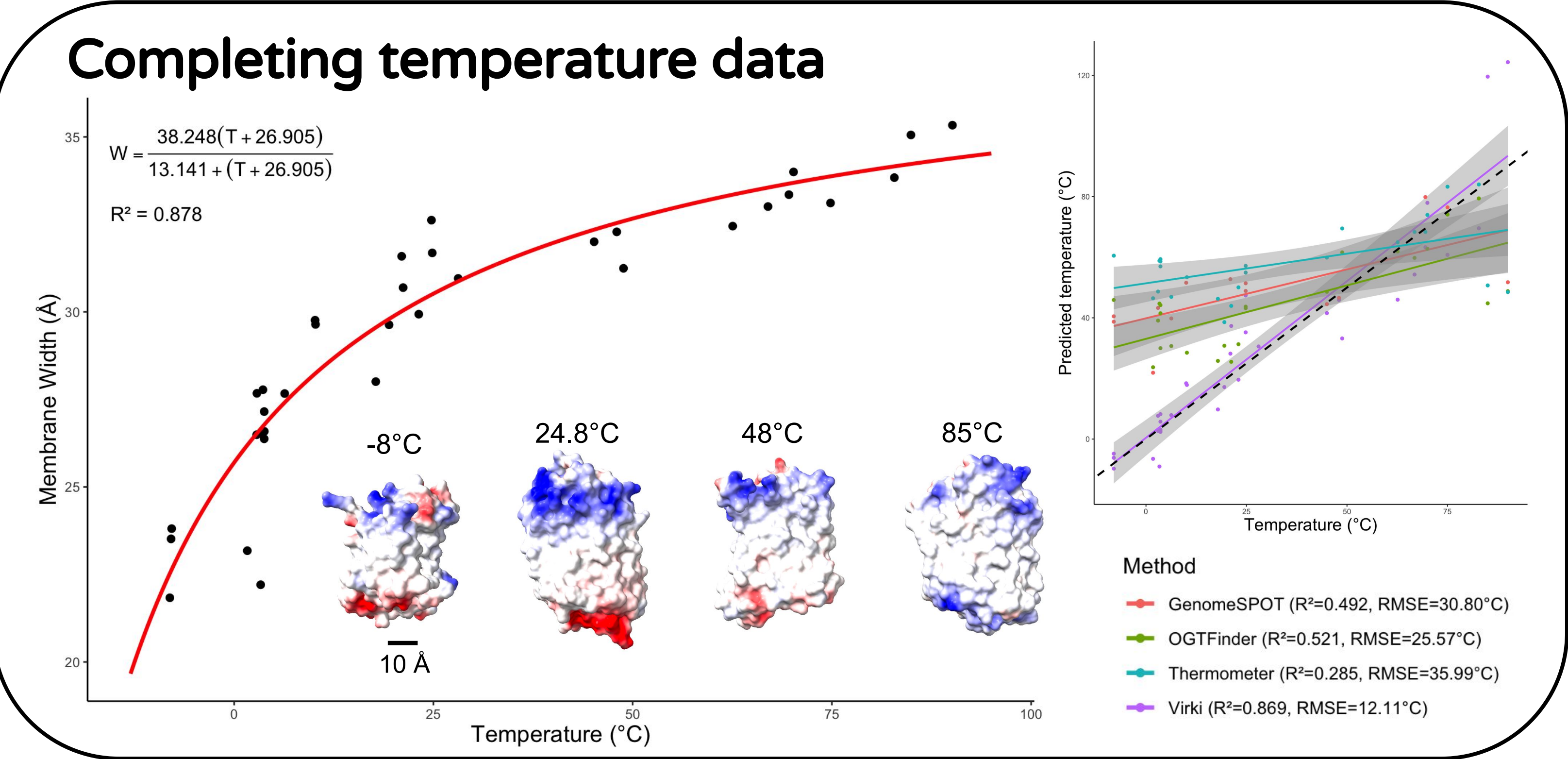
Similarity ~ Phylogeny + Geography + Environment + ϵ

Sequence, Structure

Mutations, Evolution, ...

Movement, HGT, ...

Temperature, Salinity, ...



Conclusions

- * Using biochemical properties reveals changes in sites with predicted functions and interactions.
- * Phylogeny is the main driver of the biochemical backbone and geography shows no strong effect.
- * Environment induces residue-level changes with predicted functionality.

Bibliography

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3. Panagiotou, K., Geesink, P., Köstlbacher, S. et al. Diversity, ecology, cell biology and evolution of the Asgard archaea (2026).

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